

BSCHE Graduate Written Examination – Fall 2019

Written examination conducted with graduating seniors in Chemical Engineering

Name:

Question 1

Career plans. (Scoring rubric was designed to assess potential for life-long learning.)

Where do you see yourself in 5-10 years, and what steps will you take to get there?

How have you researched your career plans and options (companies, graduate schools, PE certification, etc.)?

Does your prospective employer sponsor master's degree or MBA? Under which terms?

Question 2

Professional ethics and ethical ramifications of technology.

Can you name a recent news item or significant piece of legislation that impacts on technology or engineering, and comment on its ethical ramifications?

Have you experienced an ethical dilemma in your educational or work (e.g., co-op) experience?

How would you approach an ethical dilemma? Where or to whom would you turn for guidance? Can ethical problems be approached in terms of a systematic methodology, or are you guided more by gut reaction, personal experience, upbringing, etc.?

Are you familiar with the AIChE Code of Ethics? Can you name specific provisions in it?

Which courses at UIC have contributed most to your views on professional ethics? How?

Question 3

Working in multidisciplinary teams

Can you name two other types of engineers that you might find yourself working with?

In what context or application would you be collaborating with them?

Question 4

Numerical methods and computational proficiency (Excel, MATLAB or VBA): choose one problem.

(a) Generate a graph of the solution of the following ODE initial value problem.

$$\frac{dx}{dt} = \frac{1}{\tau} \left[x_{\text{in}}(t) - x(t) \right] - kx(t)^2, \quad x(0) = 2$$
$$x_{\text{in}}(t) = 1 + \varepsilon \sin(\omega t)$$

with the constants $\tau = 3.4$, $k = 1.2$, $\varepsilon = 0.1$ and $\omega = 5$. Can you think of a chemical engineering problem (or, at least, a course) in which such an ODE would arise?

(b) Find a numerical value for the following definite integral.

$$\int_1^7 \frac{dx}{1 + \cos^3 x + \ln x}$$

Question 5

Engineering estimate.

The waterwheel shown in the following three pictures was observed to make one complete revolution every 15 seconds. Estimate the power output (in units of horsepower)?


